

Deep Momentum: International Evidence

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Abstract

Momentum stocks have a bimodal distribution in the US market, *i.e.*, winners tend to become losers and vice versa. We investigate the existence of bimodality and the profitability of deep momentum, a machine learning return prediction model designed to address bimodality, in the UK, Japan, China, and South Korea. Bimodality turns out to be a global phenomenon and deep momentum effectively addresses bimodality and generates significant profits even in countries where momentum has little predictive power. It significantly outperforms the momentum strategy and a naïve machine learning model yielding a Sharpe ratio of about 2.0.

Keywords: Bimodality; deep momentum; machine learning; reclassification.

JEL Classification: G11, G12.

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1 Introduction

Jegadeesh and Titman (1993) document the persistence and pervasiveness of momentum in the US market. They construct a long-short portfolio that buys stocks with the highest previous returns (winners) and sells stocks with the lowest previous returns (losers) and find that the portfolio can generate significant abnormal returns for the next few months. Since its discovery, momentum has been one of the most persistent and puzzling anomalies in the finance literature, over which researchers are not yet able to reach a consensus. The predictive ability of momentum has challenged the efficient market hypothesis and has been investigated in a wide range of countries (Fama and French, 2012; Rouwenhorst, 1998, 1999). It has also been explored in a diverse array of financial assets (Moskowitz et al., 2012; Gorton et al., 2013; Asness et al., 2013). Momentum remains significant in studies that examine the factor zoo, *e.g.*, Green et al. (2017) examine 94 firm characteristics in the US market and find the price momentum is still significant. It is also found to play an important role in recent machine learning-based asset pricing studies (Madge and Bhatt, 2015; Peachavanish, 2016; Gu et al., 2020; Han, 2022).

The performance of momentum, however, is not stable, and this instability often occurs in times of market stress. Chordia and Shivakumar (2002) document that the momentum profit is statistically significantly positive in economic expansions but insignificantly negative in economic recessions, which suggests that the momentum premium is pro-cyclical. Blitz et al. (2011) find similar evidence: the return of a momentum strategy exhibits a substantial return during expansionary periods, while it suffers a negative return during recessionary periods. They also find that the losses during recessions are concentrated in the second half of the period when the market tends to revert. Daniel and Moskowitz (2016) document two major momentum crash periods in the US market - June 1932 to December 1939 and March 2009 to March 2013. During a crash period, past losers gain more than past winners, and the gap can be as large as 200%. Since momentum crashes introduce huge risks and losses to investors, how to control such risk has brought challenges to researchers. Daniel and Moskowitz (2016) propose an optimal dynamic momentum strategy based on the forecast of momentum's mean and variance. Barroso and Santa-Clara (2015) propose a constant volatility strategy, which controls the risk level of the momentum strategy.

Han (2022) documents the risk of the momentum strategy from a cross-sectional perspective and

develops a machine learning-based return prediction model that addresses momentum crashes and significantly outperforms the traditional momentum strategy. He reveals that momentum stocks have a U-shaped bimodal relative return distribution, which implies that although past winners are most likely to belong to the highest-return decile in the future, their probability of belonging to the lowest-return decile is also significant. A similar observation can be made for past losers, which makes the momentum strategy fundamentally risky. He also finds that a naïve machine learning model cannot address bimodality and proposes a novel prediction model that can alleviate the risk of the bimodal distribution. He employs machine learning to estimate the relative return distribution and reclassify stocks based on their predicted financial performance such as the expected return or Sharpe ratio. The empirical results show that the proposed model, deep momentum (DM), significantly outperforms the traditional momentum strategy as well as a naïve machine learning model in the US stock market, and a value-weighted long-short portfolio can earn an annualized mean return of 35% and a Sharpe ratio of 1.66.

Inspired by the work of Han (2022), the aim of this paper is to provide answers to the following questions. First, we examine the existence of bimodality in other countries. Examining other countries can help us better understand this phenomenon. If the bimodal return distribution of momentum stocks also exists in other countries, it will imply that the fundamental risk of the traditional momentum strategy is not only limited to the US stock market. Second, we aim to investigate the profitability of the deep momentum strategy in other countries. We choose countries where momentum strategies do not perform well and examine whether DM can still generate profits in these markets. By alleviating bimodality, DM exhibits significant predictive power and profitability in the US stock market. If bimodality is also present in other countries, DM is likely to be able to alleviate it and generate profits in those countries. Does DM perform well only in the US stock market, or can it be applied to a broader region? It will be interesting to know the answer.

We investigate the existence of bimodality and test the DM strategy in four countries: the UK, Japan, China, and South Korea. The UK is chosen because it is one of the largest stock markets and the traditional momentum strategy performs well in the UK market. We test whether DM can outperform the momentum strategy in this market. The other three countries are chosen because the momentum strategy is known to perform poorly in Asian markets and these three countries are

representative of the Asian market.

We find that bimodality exists not only in the US but also in the four countries we examine. Past winners exhibit bimodality in Japan, China, and South Korea, and past losers exhibit bimodality in all four countries. We also find supporting evidence of the DM model. The traditional momentum strategy yields annualized Sharpe ratios of 0.57, 0.00, -0.06, and 0.15, respectively in the UK, Japan, China, and South Korea. A naïve machine learning model outperforms the momentum strategy and yields Sharpe ratios of 0.94, 0.92, 0.68, and 0.98 in these countries. In contrast, a DM strategy that reclassifies stocks based on expected return further improves the performance and yields Sharpe ratios of 1.66, 1.11, 1.80, and 1.64. When size dummies are added to the predictors, the Sharpe ratio increases to 1.97, 1.75, 1.88, and 2.76. The DM strategy does not suffer any significant losses during the global financial crisis and the return of the DM strategy cannot be explained by the Fama-French three factors or the momentum factor.

We contribute to the asset pricing literature by providing supportive evidence of bimodality and demonstrating that momentum features combined via machine learning can exhibit predictive power even when a single momentum feature has little predictive power.

This paper is organized as follows. Section 2 reviews the literature on the international evidence of momentum, Section 3 describes the methodology of Han (2022), Section 4 carries out empirical analyses, and Section 5 concludes.

2 International evidence of momentum

This section reviews the literature on momentum in the four countries we examine. The literature finds supportive evidence of momentum in the UK, while the evidence appears to be weak in Asian countries.

2.1 The United Kingdom

The UK stock market is one of the world's oldest stock markets. The London Stock Exchange (LSE) is the largest in Europe ahead of Euronext, which has more than 2000 companies currently listed. Rouwenhorst (1998) examines the momentum strategy using stocks from twelve European countries, which include 494 stocks from the UK. Using the sample from 1980 to 1994, he finds

that the momentum strategy yields a significantly positive return in the UK. Lui et al. (1999) find significant momentum profits in the UK during the period from 1977 to 1998, which cannot be explained by the Fama-French three factors. Hon and Tonks (2003) also document evidence that supports this result. Agyei-Ampomah (2007) reveals that the loser portfolio is heavily weighted on small and illiquid stocks, and a momentum portfolio can incur losses if held for a short horizon when transaction costs are taken into account. Siganos (2010) constructs a momentum strategy that consists of extreme winners and losers and documents statistically significant momentum profits from 1988 to 2006.

2.2 Japan

The evidence of momentum in Japan is mixed. Asness (2011) finds it difficult to capture momentum premium in Japan. Chou et al. (2007) suggest that the Japanese market exhibits a contrarian effect rather than a momentum effect. They are unable to find a significant momentum premium in Japan for the period from 1981 to 2003. They attribute the weak momentum effect to cultural differences: Japanese are more collectivists than individualists and lack confidence, which results in a slow price reaction to information. Fama and French (2012) test size, value, and momentum premiums in the international stock markets from 1989 to 2011 and do not find significant evidence of momentum in the Japanese market. Hanauer (2014) notes that momentum returns are higher when the market is stable compared to when it transitions to another state and claims that the high transition frequency of the Japanese market leads to low momentum profits. Barroso and Santa-Clara (2015) demonstrate that the performance of momentum can be improved by managing its risk, but the Sharpe ratio is still low at 0.24 after controlling the risk.

2.3 China

The Chinese stock market began in December 1990 and has a relatively short history. Nevertheless, it has attracted significant interest from international investors, especially after A-shares became a component of the MSCI emerging market index in June 2017. The literature on momentum in the Chinese stock market is divided. Using A-shares from 1993 to 2000, Kang et al. (2002) find statistically significant intermediate-horizon momentum profits in China. Wang and Chin (2004) document that low-volume stocks exhibit momentum while high-volume stocks exhibit return

reversal, which is contrary to the US market, where momentum profits are found across all volume levels. They attribute the difference to the unique characteristics of the Chinese stock market, such as the prohibition of short selling and the dominance of inexperienced individual investors. Similarly, Cheema and Nartea (2014) argue that momentum returns are not necessarily higher for firms with greater information uncertainty due to the restrictions placed on short selling. They also report a considerable loss of momentum strategy in the Chinese market during the financial crisis. Meanwhile, Chen et al. (2018) show that a low-frequency momentum strategy is profitable in China even after controlling for size and bid-ask bounce. Yang et al. (2019) review the majority of the literature on momentum in China and find that different studies employ different filters, which may lead to designed results.

2.4 South Korea

The literature on momentum in South Korea is rare and most existing studies do not observe a significant momentum profit. Chui et al. (2000) claim that momentum does not exist in South Korea during the period from 1997 to 2000. Although momentum acquires a positive return, it is not statically significant. They suggest that this result may be caused by the high volatility of the momentum portfolio during the 1997 Asian financial crisis. Furthermore, Chui et al. (2000) report an insignificant negative momentum profit from 1978 to 1997. Chae and Eom (2009) also find negative momentum returns from 1980 to 2005, which they attribute to investors' underreaction to market-wide information. Park and Kim (2014) report a negative momentum return of -0.40% during the period from 1990 to 2010. Pyo and Yong (2013) document a positive relationship between momentum return and idiosyncratic volatility and acquire a positive momentum profit after carefully selecting stocks based on the relationship.

3 Methodology

We adopt Han (2022)'s deep momentum (DM) return prediction model for our empirical analysis. DM is a classification model that predicts a stock's future return decile and consists of two steps. The first step estimates the cross-sectional return distribution via a neural network and the second step reclassifies stocks based on their predicted financial performance. For the completeness of the

paper, the DM model is described below.

3.1 Cross-sectional return distribution estimation

Han (2022) employs two classifiers to estimate the cross-sectional return distribution: a nominal classifier and an ordinal classifier. The nominal classifier is a conventional neural network multiclass classifier that does not take the order of the classes into account. The ordinal classifier is a collection of binary classifiers and is designed to take the order into account. Return classes are ordered and misclassifying a stock into a class adjacent to the true class is less harmful than misclassifying it into a class far from the true class. In this sense, the ordinal classifier is considered a better choice for a return prediction problem. However, Han (2022) finds that the ordinal classifier does not outperform the nominal classifier, especially after reclassification. Therefore, we consider only the nominal classifier for our empirical analysis.

Let $\mathbf{x} = \{x_1, \dots, x_M\}$ denote M input features (explanatory variables), and c and $\mathbf{y}$ the target variable (output class) and its one-hot encoding, respectively: if possible outcomes are $1, \dots, K$, and the true class is k , then $c = k$ and $\mathbf{y}$ is a K -dimensional vector with the k -th element equal to 1 and the other elements equal to 0.

Given a dataset of N observations, $\{\mathbf{x}^i, \mathbf{y}^i\}_{i=1}^N$, a neural network is trained so that the following cost function is minimized:

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N L(\mathbf{y}^i, \hat{\mathbf{y}}^i(\mathbf{x}^i; \theta)), \quad (1)$$

where θ is a set of model parameters, and $\hat{\mathbf{y}}(\mathbf{x}; \theta)$ is the output of the neural network, whose k -th element is given by the softmax function:

$$\hat{y}_k(\mathbf{x}; \theta) = \frac{e^{z_k(\mathbf{x}; \theta)}}{\sum_{k=1}^K e^{z_k(\mathbf{x}; \theta)}}, \quad k = 1, \dots, K, \quad (2)$$

for some function $z_k(\mathbf{x}; \theta)$. The exact form of $z_k(\mathbf{x}; \theta)$ is determined by the network architecture. Note that $\hat{y}_k(\mathbf{x}; \theta)$ can be interpreted as the probability of a sample with features $\mathbf{x}$ belonging to

class k , $P(c = k|\mathbf{x})$. The loss function is defined as the cross-entropy

$$L(\mathbf{y}, \hat{\mathbf{y}}(\mathbf{x}; \boldsymbol{\theta})) = - \sum_{k=1}^K y_k \log \hat{y}_k(\mathbf{x}; \boldsymbol{\theta}). \quad (3)$$

3.2 Reclassification

The standard method to predict the class of a stock is to choose the class with the maximum probability, *i.e.*, $\hat{c} = \operatorname{argmax}_k \hat{y}_k$. However, the class with the highest probability does not represent the stock's expected return when the distribution is bimodal, and the stocks classified into the top (bottom) class are unlikely the ones with the highest (lowest) expected returns. To address this issue, Han (2022) proposes five reclassification methods and finds that two methods that reclassify stocks based on the predicted expected return and one based on the Sharpe ratio perform particularly well. We adopt these three methods.

Given the expected returns of the classes, μ_k , $k = 1, \dots, K$, the expected return of a stock can be obtained from the law of total expectation:

$$\mu = E[r] = E[E[r|c]] = \sum_{k=1}^K P(c = k) \mu_k. \quad (4)$$

Using the predicted probabilities from Section 3.1, the expected return can be estimated as follows:

$$\hat{\mu} = \sum_{k=1}^K \hat{y}_k \mu_k. \quad (5)$$

The expected return can then be used to reclassify stocks. Since the mean returns of the classes are unknown, they need to be estimated. The first two reclassification methods take different approaches to estimate them.

Reclassification on Probability Difference (PrDf) The first reclassification method assumes that the mean returns of the classes decrease linearly from top to bottom. Under this assumption, it can be shown that the mean return of a stock is proportional to

$$PrDf = \sum_{k=1}^{K/2} (\hat{y}_k - \hat{y}_{K+1-k}) \left(\frac{K}{2} + 1 - k \right). \quad (6)$$

PrDf reclassifies stocks based on this value.

Reclassification on Mean Return (Return) The second method estimates the mean returns of the classes using the sample analogue over the past five years. The return of class k is defined as the average return of the stocks that belong to the class. For instance, the return of class 0 is the average return of top 10% stocks in terms of return. Denoting the sample mean of class k 's return $\hat{\mu}_k$, the estimate of a stock's mean return can be obtained from Equation (4):

$$\hat{\mu} = \sum_{k=1}^K \hat{y}_k \hat{\mu}_k. \quad (7)$$

Reclassification on Sharpe Ratio (Sharpe) The last method reclassifies stocks based on the Sharpe ratio. Using the law of total variance, the variance of stock return is given by

$$\sigma^2 = V[r] = E[V[r|c]] + V[E[r|c]] = \sum_{k=1}^K P(k) (\sigma_k^2 + \mu_k^2) - \mu^2, \quad (8)$$

where σ_k^2 is the variance of class k 's return. Substituting μ_k and σ_k^2 with their sample analogues $\hat{\mu}_k$ and $\hat{\sigma}_k^2$, the variance of a stock's return can be estimated using the equation

$$\hat{\sigma}^2 = \sum_{k=1}^K \hat{y}_k (\hat{\sigma}_k^2 + \hat{\mu}_k^2) - \hat{\mu}^2. \quad (9)$$

The Sharpe ratio of the stock is then estimated by $\hat{SR} = \hat{\mu}/\hat{\sigma}$.

3.3 Input Features

Following Han (2022), we choose five momentum features as the input features of the neural network. The m -month price momentum, MOM_m , is defined as the cumulative return over the period from $t - m$ to $t - 2$ for $m = 3, 6, 9, 12$, and the previous one-month return for $m = 1$:

$$MOM_m = \prod_{j=t-m}^{t-2} (r_j + 1) - 1, \quad m = 3, 6, 9, 12, \quad (10)$$

$$MOM_1 = r_{t-1}, \quad m = 1, \quad (11)$$

where r_j denotes the return in month j . To eliminate the market trend, these are normalized cross-sectionally in each month using the formula

$$nMOM_m = \frac{MOM_m - M_{MOM_m}}{S_{MOM_m}}, \quad (12)$$

where M_{MOM_m} and S_{MOM_m} respectively denote the cross-sectional mean and standard deviation of MOM_m . The final feature set consists of the normalized momentum features, $nMOM_m$, and the cross-sectional means, M_{MOM_m} . The latter is included to take the macroeconomic status into account.

Han (2022) also includes a size feature and finds that the size feature can significantly improve the performance of the strategy. The size feature is defined as follows: in each month, stocks are divided into deciles on the market capitalization at the end of the previous month and assigned one of ten size dummies, D_s , $s = 1, \dots, 10$. We use only momentum features for our primary empirical analysis as our objective is to compare the DM model with the traditional momentum strategy. The results that include size dummies are discussed after the discussion of the primary results.

3.4 Neural network architecture and hyperparameter tuning

We test a different number of layers between 3 and 10 and a different number of neurons in (12, 24, 36, 64, 128), and choose the final architecture via cross-validation using the data before 2000. The best architecture turns out to be one with 3 or 4 layers and 36 or 64 neurons per layer depending on the sample. Since the prediction performance is similar among these choices, we choose an architecture with 3 layers and 64 neurons for all the countries. We use the rectified linear unit (ReLU) as the activation function of a hidden layer and the Adam optimizer. The number of epochs is determined via early stopping.

3.5 Model Evaluation

The model is evaluated using conventional classification performance metrics. For the overall classification performance, the average loss and accuracy are employed:

$$\text{Average Loss} = \frac{1}{N} \sum_{i=1}^N L(\mathbf{y}^i, \hat{\mathbf{y}}^i), \quad (13)$$

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{N}. \quad (14)$$

To assess the performance in each class, the following metrics are employed:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (15)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (16)$$

$$\text{F1-score} = \frac{2}{1/\text{Precision} + 1/\text{Recall}}, \quad (17)$$

where TP , FP , and FN respectively denote true positive, false positive, and false negative. Precision measures how many of the samples predicted to be true are actually true and are deemed more relevant to return prediction than recall: missing some stocks that are going to have a high or low return is not harmful, but misclassifying a low-return stock into a high-return class or vice versa can be disastrous.

The following error measures designed to account for ordinality are also employed:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (c^i - k)^2 \hat{y}_k^i, \quad (18)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K |c^i - k| \hat{y}_k^i. \quad (19)$$

Both MSE and MAE have their minimum value of 0 when the predicted probability for the true class is 1, and increase as the probabilities for distant classes increase. These measures can also be defined at the class level: they are respectively denoted by MSEP and MAEP when stocks are grouped on predicted classes, and by MSER and MAER when grouped on true classes. MSEP and MAEP are similar to precision in the sense that they measure the prediction error within a

predicted class, whereas MSER and MAER measure the prediction error within a true class and are similar to recall. Like precision, MSEP and MAEP are deemed more relevant to stock classification.

4 Empirical Analysis

4.1 Data

We collect from Datastream monthly returns, prices, volumes, and market capitalizations of the stocks listed in four countries: the UK, Japan, China, and South Korea. The UK sample consists of all common shares listed on the London Stock Exchange, and the sample period is from January 1965 to December 2019. The Japan sample consists of all common shares listed on the Tokyo Stock Exchange and the Osaka Securities Exchange, and the sample period is from January 1973 to December 2019. The China sample consists of A shares listed on the Shanghai Stock Exchange, and the sample period is from December 1990 to December 2019. Finally, the Korea sample consists of all common shares listed on the Korea Stock Exchange, and the sample period is from July 1996 to December 2019. We include both listed and delisted stocks to avoid survivorship bias.

We set the test period to be from January 2000 to December 2019 for all countries and re-train the model every year during the test period. For the UK and Japan samples, we use the past 240 months of data to train the model in January 2000 and expand the sample every year. For the China and Korea samples, which do not have 20 years of data in January 2000, we use all available historical data on the training date. The training sample is randomly split into a training set and a validation set in a ratio of 7:3.

4.2 Classification Performance

This section evaluates the classification performance of the DM model. Table 1 presents the overall classification performance in each market. The test set accuracy and validation set accuracy are comparable to the accuracy in the training set in all countries, which suggests that the model does not suffer from overfitting. The accuracy in the test is above 14% in all countries except for China, where the accuracy is slightly lower at 13.58%. As a reference, the test set accuracy in the US market reported in Han (2022) is 14.8%. These values are significantly higher than the accuracy of random guess, 10%. The model performs particularly well in the UK, where the test set accuracy

is 14.84% and the MSE is 14.63. As shown later, the traditional momentum strategy performs well in the UK, which implies that the momentum features have more predictive power compared to the other countries.

Table 1: Classification performance

This table reports the classification performance of DM. The evaluation metrics are defined in Section 3.5. The training set (Train) and validation set (Valid) metrics are the averages of the metrics obtained from yearly retraining, and the test set (Test) metrics are obtained from the test period, 2000.01 to 2019.12.

	Train				Valid				Test			
	Loss	Acc	MSE	MAE	Loss	Acc	MSE	MAE	Loss	Acc	MSE	MAE
UK	2.24	14.86	14.99	3.10	2.25	13.99	15.10	3.11	2.22	14.84	14.63	3.05
Japan	2.26	14.35	16.01	3.21	2.26	13.97	16.09	3.22	2.26	14.12	16.00	3.21
China	2.26	14.64	16.05	3.23	2.27	13.54	16.18	3.24	2.27	13.58	16.06	3.23
S. Korea	2.24	16.04	15.79	3.18	2.26	14.08	15.99	3.21	2.26	14.04	16.07	3.21

Table 2 compares the classification performances in the highest-return decile (H) and the lowest-return decile (L) before and after reclassification. Before reclassification (Org), the precisions of both H and L are higher than the average (avg) in all countries except for South Korea, where the precision of H is slightly lower than the average. The precision of L is particularly high exceeding 20% in most countries. The table also shows that the model tends to classify more stocks into L and fewer stocks into H. These results suggest that predicting losers is easier and are consistent with the empirical fact that the profit of the momentum strategy mainly results from the loser decile.

When stocks are reclassified, the precision of H tends to decrease, whereas the precision of L tends to increase. This result is partly because more (fewer) stocks are classified into H (L) after reclassification. In contrast, MSEP and MAEP, the error metrics that take the ordinality into account, generally decrease in both classes after reclassification, and the error reduction is particularly evident in H. For instance, the MSEP of H in the UK decreases from 21.25 to 11.00 after reclassification on the Sharpe ratio, which is a stark contrast to the precision that decreases from 16.63 to 5.45. The reduction of both precision and MSEP implies that some stocks that are correctly classified into H are missed after reclassification while many low-return stocks misclassified into H are replaced by higher-return stocks. From this result, we can expect that reclassification is likely to improve the financial performance of H more significantly than that of L. Reclassification can effectively eliminate some stocks with opposite returns and group stocks with more similar

returns. The opposite changes of precision and MSEP reveal that conventional evaluation metrics such as precision can be misleading and making an investment decision based on them can result in unexpectedly poor financial performance.

To further examine the effect of reclassification, we draw the predicted probabilities and realized distributions of the stocks in H and L in Figure 1. Orange bars represent predicted probabilities and blue bars represent actual distributions of the stocks. For instance, a bar chart labeled ‘Org (H)’ presents the average predicted probability for each class of the stocks that are classified into H, and their actual distribution. From the results before reclassification, it is evident that the stocks in H and L exhibit bimodal distributions. The stocks classified into H (L) have a high probability of L (H), and many of these stocks actually generate low (high) returns and end up in L (H). The predicted distribution and actual distribution are surprisingly similar, which suggests that stocks can be better classified from a financial perspective by exploiting the distribution instead of choosing the class with the highest probability.

After stocks are reclassified, we no longer observe bimodality in H. Some correctly classified stocks are missed, but, at the same time, many stocks with a high probability of low return are eliminated. As a result, both predicted and actual distributions have a right-skewed bell shape or decrease monotonically towards low-return classes. Reclassification can also alleviate the bimodality in L but cannot eliminate it completely. Overall, these results suggest that stocks with a high probability of crash also have a high probability of jump, and it is difficult to distinguish crashing stocks from jumping stocks. These findings are similar to those from the US as reported by Han (2022).

The distributions of the traditional momentum strategy (JT) reveal that many winners turn out to yield the lowest returns and many losers yield the highest returns. In fact, there are more stocks in H that end up in L rather than H in Japan, China, and South Korea. The only exception is the UK, where the stocks in H do not exhibit bimodality.

4.3 Financial Performance

Table 3 reports the mean returns and the Sharpe ratios of equal-weighted and value-weighted portfolios. Overall, the DM model significantly outperforms the traditional momentum strategy in all countries even before reclassification. When stocks are equally weighted, the mean return

Table 2: Reclassification performance

This table reports the class-level test set classification performance of DM before and after reclassification. ‘Org’ denotes DM before reclassification, and ‘PrDf’, ‘Return’, and ‘Sharpe’ denote reclassification on probability difference, mean return, and the Sharpe ratio, respectively.

		Precision	Recall	f1-score	MSEP	MSER	MAEP	MAER	Support	Predicted	(%)
UK											
Org	H	16.63	10.23	12.67	21.25	29.08	3.70	4.51	16456	10121	5.15
	L	20.41	45.12	28.11	22.17	28.45	3.75	4.34	14740	32578	16.58
	avg	14.11	15.42	13.67	14.93	15.51	3.09	3.13	196508	196508	100.00
PrDf	H	7.01	8.41	7.64	12.35	29.08	2.81	4.51	16456	19752	10.05
	L	20.59	27.28	23.47	21.63	28.45	3.69	4.34	14740	19532	9.94
	avg	12.50	12.86	12.61	14.64	15.51	3.05	3.13	196508	196508	100.00
Return	H	10.50	12.60	11.46	15.09	29.08	3.10	4.51	16456	19752	10.05
	L	19.37	25.67	22.08	20.94	28.45	3.64	4.34	14740	19532	9.94
	avg	12.84	13.21	12.95	14.64	15.51	3.05	3.13	196508	196508	100.00
Sharpe	H	5.45	6.54	5.95	11.00	29.08	2.66	4.51	16456	19752	10.05
	L	18.85	24.98	21.49	20.54	28.45	3.61	4.34	14740	19532	9.94
	avg	11.88	12.17	11.96	14.64	15.51	3.05	3.13	196508	196508	100.00
Japan											
Org	H	15.63	15.38	15.51	19.75	29.28	3.59	4.52	63812	62805	9.78
	L	21.32	39.86	27.78	21.88	28.42	3.73	4.36	62354	116579	18.15
	avg	13.14	14.22	12.75	15.60	16.05	3.18	3.21	642235	642235	100.00
PrDf	H	9.50	9.58	9.54	14.41	29.28	3.05	4.52	63812	64330	10.02
	L	23.90	24.58	24.24	22.03	28.42	3.72	4.36	62354	64109	9.98
	avg	12.04	12.08	12.06	16.01	16.05	3.21	3.21	642235	642235	100.00
Return	H	12.12	12.22	12.17	16.71	29.28	3.28	4.52	63812	64330	10.02
	L	22.20	22.82	22.50	21.18	28.42	3.65	4.36	62354	64109	9.98
	avg	12.29	12.32	12.31	16.01	16.05	3.21	3.21	642235	642235	100.00
Sharpe	H	9.11	9.18	9.15	14.07	29.28	3.01	4.52	63812	64330	10.02
	L	21.31	21.91	21.61	20.75	28.42	3.61	4.36	62354	64109	9.98
	avg	11.65	11.68	11.66	16.01	16.05	3.21	3.21	642235	642235	100.00
China											
Org	H	13.38	13.63	13.51	18.04	28.63	3.43	4.49	16553	16869	9.83
	L	17.68	47.47	25.76	20.01	28.38	3.61	4.39	15879	42642	24.86
	avg	12.56	13.88	11.81	15.40	16.25	3.17	3.25	171550	171550	100.00
PrDf	H	10.12	10.55	10.33	14.42	28.63	3.06	4.49	16553	17258	10.06
	L	19.08	20.49	19.76	19.66	28.38	3.56	4.39	15879	17050	9.94
	avg	11.47	11.54	11.50	16.06	16.25	3.23	3.25	171550	171550	100.00
Return	H	11.21	11.68	11.44	15.55	28.63	3.17	4.49	16553	17258	10.06
	L	18.06	19.39	18.70	19.17	28.38	3.51	4.39	15879	17050	9.94
	avg	11.63	11.70	11.66	16.06	16.25	3.23	3.25	171550	171550	100.00
Sharpe	H	10.12	10.55	10.33	14.34	28.63	3.05	4.49	16553	17258	10.06
	L	18.85	20.24	19.52	19.54	28.38	3.55	4.39	15879	17050	9.94
	avg	11.41	11.48	11.44	16.06	16.25	3.23	3.25	171550	171550	100.00
South Korea											
Org	H	12.74	10.44	11.48	18.66	29.58	3.49	4.56	17370	14239	8.11
	L	20.83	46.17	28.70	21.44	27.67	3.69	4.24	16865	37384	21.30
	avg	12.92	14.17	12.78	15.61	16.15	3.18	3.22	175480	175480	100.00
PrDf	H	7.98	8.11	8.04	12.85	29.58	2.88	4.56	17370	17659	10.06
	L	25.48	26.34	25.90	22.25	27.67	3.72	4.24	16865	17439	9.94
	avg	12.04	12.09	12.06	16.08	16.15	3.22	3.22	175480	175480	100.00
Return	H	11.66	11.85	11.76	16.79	29.58	3.30	4.56	17370	17659	10.06
	L	24.14	24.96	24.54	21.58	27.67	3.67	4.24	16865	17439	9.94
	avg	12.63	12.68	12.65	16.07	16.15	3.21	3.22	175480	175480	100.00
Sharpe	H	7.98	8.11	8.04	12.74	29.58	2.87	4.56	17370	17659	10.06
	L	24.45	25.28	24.86	21.69	27.67	3.67	4.24	16865	17439	9.94
	avg	11.93	11.97	11.95	16.08	16.15	3.22	3.22	175480	175480	100.00

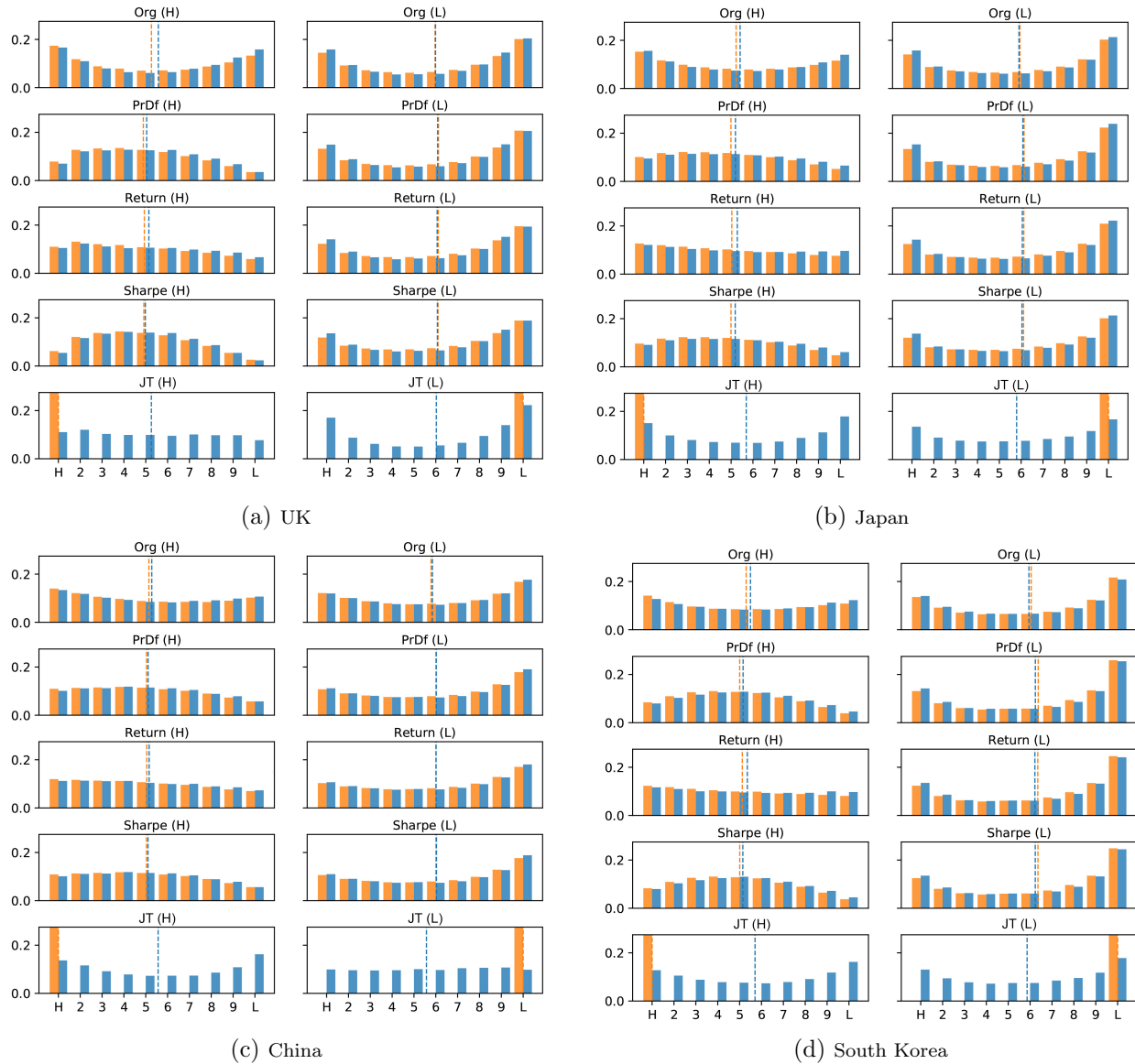


Figure 1: Cross-sectional distribution

This figure presents the cross-sectional relative return distributions in the high-return (H) and low-return (L) deciles. The orange bars represent the average predicted probabilities of the stocks and the blue bars represent their realized frequencies. The dotted lines represent the average return class. The title of each graph refers to the reclassification method, and the horizontal axis denotes the class of the one-month-ahead return.

increases as much as 28% and the Sharpe ratio increases up to 0.92. Reclassification further improves the performance by adding as much as 20% to the mean return and 1.21 to the Sharpe ratio. When stocks are value-weighted, the improvements are less pronounced but still significant. Reclassifying stocks on Return doubles the mean return and the Sharpe ratio in some countries. More discussions of the model's performance in each country are provided below.

In the UK, the momentum strategy performs well with an annualized mean return of 18% and a Sharpe ratio of 0.57. The long portfolio performs particularly well and renders an annualized mean return of 16% and a Sharpe ratio of 0.84. This result has been anticipated from Figure 1, which shows that the winners do not have a bimodal distribution. Nevertheless, DM outperforms JT significantly even before reclassification yielding an annualized mean return of 20% and a Sharpe ratio of 0.94. The greater increase in the Sharpe ratio suggests that the long-short portfolio constructed by DM is less volatile. Reclassifying stocks on Return further improves the performance and increases the mean return and the Sharpe ratio to 36% and 1.66, respectively. The other reclassification methods also improve the performance to a similar level. The value-weight portfolio performs worse. DM yields a mean return of 18% and a Sharpe ratio of 0.53 when stocks are reclassified on Return. In contrast, JT is not affected much by the weighting scheme and yields a mean return of 19% and a Sharpe ratio of 0.43. The relatively poor performance of the value-weighted portfolio is consistent with the findings in the US (Han, 2022) and can be partially attributed to the training process of the neural network, in which each stock receives the same weight in the cost function regardless of its size. As discussed in Section 4.5, adding size dummies improves the performance of both equal-weight and value-weighted portfolios significantly.

Previous literature suggests that the momentum strategy shows weak performance in the East Asian markets. Our results are consistent with the previous findings. The Sharpe ratios of the momentum strategy in Japan, China, and South Korea are close to zero, which implies that one-year momentum has little predictive power in these countries. Nevertheless, momentum features of different periods can have significant predictive power when combined together via a neural network. In Japan, DM (Org) is able to generate an annualized mean return of 9% and a Sharpe ratio of 0.92, and DM (Return) yields a mean return of 17% and a Sharpe ratio of 1.11. These values are significantly higher than those of JT, which are 0% and 0.00, respectively. In China, JT yields a mean return of -1% and a Sharpe ratio of -0.06. In contrast, DM (Org) yields a mean return

of 12% and a Sharpe ratio of 0.68, and DM (Return) yields a mean return of 27% and a Sharpe ratio of 1.80. DM (Sharpe) performs best in China with a mean return of 30% and a Sharpe ratio of 1.89. In South Korea, JT yields a mean return of 5% and a Sharpe ratio of 0.15. In contrast, DM (Org) yields a mean return of 33% and a Sharpe ratio of 0.98, and DM (Return) yields a mean return of 49% and a Sharpe ratio of 1.64.

Unlike the US and the UK, JT performs better in Asian countries when stocks are value-weighted. The mean return and the Sharpe ratio of JT are respectively 2% and 0.09 in Japan, 3% and 0.10 in China, and 6% and 0.19 in South Korea. DM performs worse when stocks are value-weighted, but still outperforms JT. The mean return and the Sharpe ratio of DM (Return) are respectively 13% and 0.72 in Japan, 25% and 1.22 in China, and 41% and 1.20 in South Korea. Among the three countries, DM performs best in South Korea in terms of mean return, while it performs best in China in terms of the Sharpe ratio.

Figure 2 presents the cumulative returns of the long-short portfolios and their legs. The figure highlights the effectiveness of reclassification. While DM outperforms JT before reclassification, it still suffers significant drawdowns at times. For instance, the long-short portfolio constructed by DM (Org) experiences a huge drop in China in 2007. In contrast, the long-short portfolio does not suffer any significant loss when stocks are reclassified. It is also notable that the return of the long-short portfolio grows steadily throughout the test period in all countries when stocks are reclassified.

4.4 Factor Regression

To examine whether the return of the DM long-short portfolio can be explained by factors, we run a factor regression using the Fama-French three factors plus the momentum factor. We use the Fama-French three factors rather than the more recent five factors as the other two factors are difficult to obtain for other countries. The UK factors are obtained from a data source at Exeter University, the Japan factors are obtained from the K. French website, the China factors are obtained from the CSMAR database, and the Korea factors are obtained from Handa partners. The sample period is from 2000 to 2019 except for the UK, for which the sample period ends in 2017 due to the availability of the factors. The portfolios constructed by DM (Return) are used for the regression analysis.

Table 3: Financial performance

This table reports the annualized mean returns and Sharpe ratios of the long (H), short (L), and long-short (H-L) portfolios constructed by the DM model and the traditional momentum strategy (JT). The returns are calculated over the test period from 2000.01 to 2019.12.

	Mean return					Sharpe ratio				
	Org	PrDf	Return	Sharpe	JT	Org	PrDf	Return	Sharpe	JT
UK										
H	0.16	0.20	0.23	0.20	0.16	0.48	0.98	0.92	1.07	0.84
L	-0.05	-0.12	-0.13	-0.15	-0.01	-0.15	-0.42	-0.48	-0.58	-0.04
H-L	0.20	0.32	0.36	0.35	0.18	0.94	1.42	1.66	1.56	0.57
Japan										
H	0.17	0.18	0.19	0.18	0.10	0.63	0.94	0.84	0.96	0.41
L	0.08	0.03	0.02	0.02	0.10	0.29	0.11	0.09	0.09	0.37
H-L	0.09	0.15	0.17	0.16	0.00	0.92	0.93	1.11	1.03	0.00
China										
H	0.15	0.27	0.26	0.27	0.11	0.46	0.73	0.71	0.74	0.35
L	0.03	-0.02	-0.02	-0.02	0.13	0.09	-0.05	-0.05	-0.07	0.34
H-L	0.12	0.29	0.27	0.30	-0.01	0.68	1.84	1.80	1.89	-0.06
South Korea										
H	0.22	0.27	0.25	0.27	0.03	0.47	0.60	0.57	0.63	0.09
L	-0.12	-0.25	-0.24	-0.25	-0.02	-0.32	-0.69	-0.67	-0.70	-0.04
H-L	0.33	0.52	0.49	0.52	0.05	0.98	1.46	1.64	1.64	0.15

(a) Equal-weighted portfolios

	Mean return					Sharpe ratio				
	Org	PrDf	Return	Sharpe	JT	Org	PrDf	Return	Sharpe	JT
UK										
H	0.12	0.09	0.14	0.09	0.10	0.32	0.42	0.57	0.49	0.47
L	-0.06	-0.09	-0.04	-0.07	-0.08	-0.15	-0.24	-0.12	-0.21	-0.19
H-L	0.18	0.18	0.18	0.16	0.19	0.53	0.54	0.53	0.49	0.43
Japan										
H	0.09	0.12	0.11	0.12	0.05	0.32	0.55	0.49	0.55	0.22
L	0.03	-0.02	-0.02	-0.02	0.03	0.12	-0.07	-0.07	-0.08	0.10
H-L	0.06	0.14	0.13	0.14	0.02	0.36	0.70	0.72	0.73	0.09
China										
H	0.15	0.24	0.22	0.24	0.11	0.49	0.73	0.65	0.72	0.37
L	0.02	-0.03	-0.03	-0.03	0.09	0.07	-0.10	-0.10	-0.11	0.24
H-L	0.13	0.27	0.25	0.27	0.03	0.58	1.32	1.22	1.32	0.10
South Korea										
H	0.06	0.11	0.11	0.08	-0.07	0.13	0.24	0.23	0.17	-0.17
L	-0.17	-0.28	-0.31	-0.30	-0.13	-0.46	-0.70	-0.79	-0.79	-0.27
H-L	0.23	0.39	0.41	0.39	0.06	0.64	0.93	1.20	1.02	0.19

(b) Value-weighted portfolios

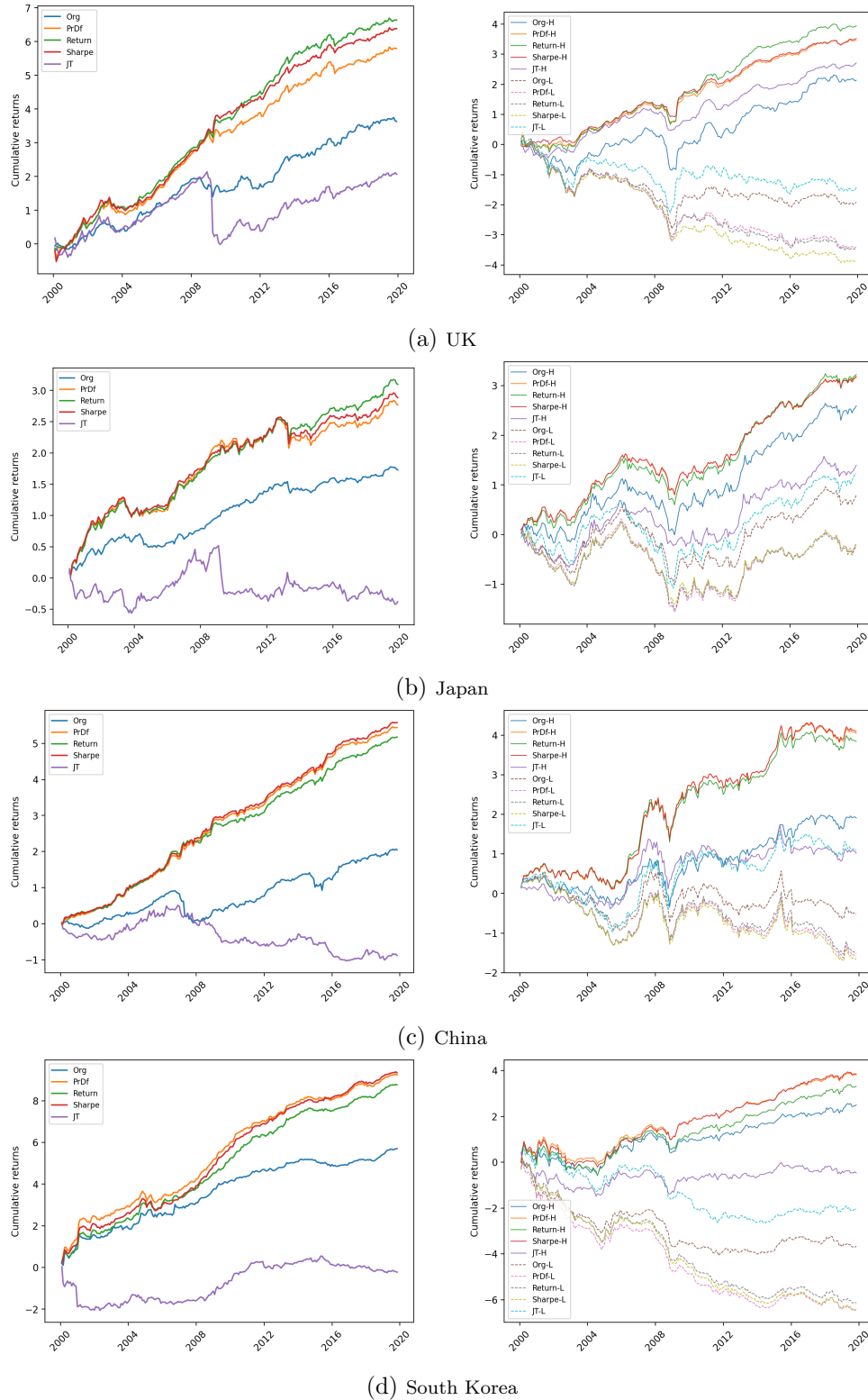


Figure 2: Cumulative Returns (equal-weighted)

This figure presents the cumulative returns (logarithmic scale) of the equal-weighted long-short portfolios constructed by DM and the traditional momentum strategy (JT). The left panels present the long-short portfolios and the right panels present long (solid line) and short (dashed line) portfolios separately.

Table 4 reports the regression results. When stocks are equally weighted, the long-short portfolio alpha is statistically and economically significant with the t -statistic greater than 6.0 in all countries but Japan, where the t -statistic is 4.38. When stocks are value-weighted, the t -statistic of alpha is greater than 4.0 in China and South Korea, 2.70 in Japan, and 1.95 in the UK. As discussed later, when size dummies are included in the input feature set, the alpha of the value-weighted long-short portfolio becomes statistically and economically significant in all countries with the t -statistic greater than 3.0. This result suggests that the return of the DM long-short portfolio cannot be explained by the Fama-French three factors or the momentum factor.

Table 4: Factor regression

This table reports the factor regression results of the long (H), short (L), and long-short (H-L) portfolios constructed by DM (Return). The portfolio returns are regressed on the Fama-French three factors (Mkt-Rf, SMB, HML) plus the momentum factor (MOM). For each portfolio, the first row reports the coefficients and the second row reports the Newey-West adjusted t -statistics. The sample period is from 2000 to 2019 except for the UK, for which the sample period ends in 2017 due to data availability.

		α	Mkt-Rf	SMB	HML	MOM	α	Mkt-Rf	SMB	HML	MOM
		Equal-weighted					Value-weighted				
UK	H	0.019	0.074	0.330	-0.181	-0.051	0.010	0.146	0.341	-0.040	-0.036
		3.54	0.54	2.16	-1.03	-0.45	1.87	1.09	2.27	-0.23	-0.33
	L	-0.012	0.109	0.590	-0.268	-0.206	-0.007	0.361	0.655	-0.053	-0.178
		-2.14	0.75	3.63	-1.43	-1.70	-0.92	2.01	3.24	-0.23	-1.18
	H-L	0.029	-0.027	-0.258	0.077	0.153	0.014	-0.208	-0.311	0.003	0.139
		6.36	-0.23	-1.97	0.51	1.58	1.95	-1.11	-1.48	0.01	0.89
Japan	H	0.014	0.136	0.330	-0.098	-0.075	0.008	0.195	0.257	-0.069	-0.120
		3.17	1.43	1.94	-0.64	-1.01	1.69	2.00	1.47	-0.44	-1.57
	L	0.000	0.099	0.469	-0.175	-0.163	-0.003	0.181	0.444	-0.223	-0.133
		-0.06	0.92	2.45	-1.01	-1.94	-0.73	1.78	2.45	-1.36	-1.68
	H-L	0.013	0.042	-0.132	0.067	0.085	0.010	0.019	-0.180	0.144	0.011
		4.38	0.65	-1.16	0.65	1.71	2.70	0.25	-1.29	1.15	0.18
China	H	0.019	0.119	0.344	-0.087	0.061	0.017	0.125	0.232	-0.246	0.091
		2.57	1.26	1.98	-0.32	0.55	2.49	1.45	1.46	-1.00	0.91
	L	-0.002	0.113	0.138	-0.218	0.049	-0.003	0.172	0.055	-0.223	0.050
		-0.27	1.24	0.82	-0.84	0.46	-0.48	2.05	0.36	-0.93	0.51
	H-L	0.019	0.007	0.205	0.131	0.012	0.018	-0.046	0.176	-0.023	0.042
		6.08	0.18	2.82	1.16	0.26	4.24	-0.85	1.77	-0.15	0.66
S. Korea	H	0.015	1.566	1.648	-0.576	-0.160	0.003	1.673	1.337	-0.532	-0.136
		2.78	14.82	11.52	-3.65	-0.82	0.48	13.54	7.99	-2.88	-0.60
	L	-0.026	1.312	1.384	-0.573	-0.169	-0.031	1.344	1.309	-0.678	-0.206
		-5.63	15.02	11.69	-4.39	-1.05	-5.71	13.05	9.39	-4.41	-1.09
	H-L	0.038	0.256	0.263	-0.004	0.007	0.031	0.332	0.026	0.146	0.067
		6.67	2.34	1.77	-0.02	0.04	4.76	2.68	0.15	0.79	0.30

4.5 Inclusion of size dummies

When size dummies are included in the input feature set, all evaluation metrics in the test set improve in most countries as shown in Table 5. The effects of the size dummies become more evident when we compare the financial performance. Table 6 reports the annualized mean returns and the Sharpe ratios of the portfolios constructed by DM including size dummies. In the UK, the mean return of the long-short portfolio increases from 20% to 24% and the Sharpe ratio increases from 0.94 to 1.01 before reclassification compared to the DM without size dummies. When stocks are reclassified on Return, these values increase from 36% to 38% and from 1.66 to 1.97, respectively. In Japan, the mean return of the long-short portfolio increases from 9% to 15% and the Sharpe ratio increases from 0.92 to 1.56 before reclassification. When stocks are reclassified on Return, these values increase from 17% to 24% and from 1.11 to 1.75, respectively. In China, the mean return of the long-short portfolio remains the same at 12% and the Sharpe ratio decreases from 0.68 to 0.58 before reclassification. However, when stocks are reclassified on Return, these values increase from 27% to 31% and from 1.80 to 1.88, respectively. Reclassifying stocks on Sharpe renders the best performance in China with a mean return of 33% and a Sharpe ratio of 1.95. In South Korea, the mean return of the long-short portfolio increases from 33% to 42% and the Sharpe ratio increases from 0.98 to 1.70 before reclassification. When stocks are reclassified on Return, these values increase from 49% to 78% and from 1.64 to 2.76, respectively. The performance of DM is remarkable especially when we consider that the momentum strategy performs poorly in Asian countries.

Size dummies improve the performance of the value-weighted portfolios as well, especially in the UK and South Korea. When stocks are reclassified on Return, the mean return and the Sharpe ratio respectively increase from 18% to 32% and from 0.53 to 1.09 in the UK, from 13% to 15% and from 0.72 to 0.78 in Japan, from 25% to 29% and from 1.22 to 1.29 in China, and from 41% to 55% and from 1.20 to 1.88 in South Korea. It is notable that size dummies do not always add value in DM when stocks are not reclassified. In the UK, the Sharpe ratio increases only slightly from 0.53 to 0.66, and in China, it decreases from 0.58 to 0.20. This result highlights the importance of reclassifying stocks on their predicted financial performance.

The factor regression results in Table 7 reveal that the returns of the long-short portfolios cannot

Table 5: Classification performance: including size

This table reports the classification performance of DM including size dummies. The evaluation metrics are defined in Section 3.5. The training set (Train) and validation set (Valid) metrics are the averages of the metrics obtained from yearly retraining, and the test set (Test) metrics are obtained from the test period, 2000.01 to 2019.12.

	Train				Valid				Test			
	Loss	Acc	MSE	MAE	Loss	Acc	MSE	MAE	Loss	Acc	MSE	MAE
UK	2.23	15.41	14.83	3.07	2.24	14.41	14.99	3.09	2.22	15.02	14.59	3.04
Japan	2.24	15.26	15.72	3.17	2.26	14.08	16.01	3.21	2.26	14.22	15.96	3.20
China	2.27	14.23	16.10	3.24	2.27	13.41	16.18	3.25	2.27	13.65	16.07	3.24
S. Korea	2.23	16.30	15.70	3.17	2.26	14.05	15.84	3.19	2.26	14.16	15.99	3.20

be explained by the factors and confirm the important role of size dummies. The t -statistic of the long-short portfolio alpha increases in all countries. When stocks are equally weighted, it is greater than 6 and as high as 11.71 (South Korea), and when stocks are value-weighted, it is greater than 3 and as high as 7.81 (South Korea).

The significant difference in performance between equal-weighted portfolios and value-weighted portfolios and the important role of size dummies presented above suggest that the predictive power of the momentum features varies across the size of the firm. In fact, it is well known that a large portion of the performance of the momentum strategy is derived from small stocks. To examine further the effect of size on the performance of DM and JT, we draw heatmaps in Figure 3. In each heatmap, the horizontal axis represents the size decile (small to big), the vertical axis represents the predicted class (low to high) by a model, and the colour of a cell represents the average return of the stocks in the cell. ‘Return’ and ‘Return (Size)’ on the vertical axis respectively refer to the DM (Return) model with and without size dummies, and ‘JT’ refers to the traditional momentum strategy.

The figure clearly shows that the momentum strategy (JT) cannot distinguish winners from losers within each size decile in all countries. We rather observe the reversal of the return among the largest firms, *i.e.*, past winners have lower returns than past losers. The positive return of the momentum strategy in the UK and South Korea appears to be mainly due to the selection of stocks of different sizes, not due to its ability to distinguish winners from losers within each size decile.

In contrast, DM has a better ability to distinguish winners from losers as evidenced by more red colours at the top and more blue colours at the bottom. DM also exhibits return reversal among large firms, but the degree of reversal is less severe compared to JT. When size dummies are added

Table 6: Financial performance: including size

This table reports the annualized mean returns and Sharpe ratios of the long (H), short (L), and long-short (H-L) portfolios constructed by the DM model including size dummies and the traditional momentum strategy (JT). The returns are calculated over the test period from 2000.01 to 2019.12.

	Mean return					Sharpe ratio				
	Org	PrDf	Return	Sharpe	JT	Org	PrDf	Return	Sharpe	JT
UK										
H	0.16	0.19	0.22	0.19	0.16	0.47	1.01	0.91	1.23	0.84
L	-0.08	-0.15	-0.16	-0.15	-0.01	-0.26	-0.53	-0.60	-0.62	-0.04
H-L	0.24	0.34	0.38	0.34	0.18	1.01	1.66	1.97	1.79	0.57
Japan										
H	0.23	0.23	0.25	0.23	0.10	0.93	1.14	1.11	1.16	0.41
L	0.09	0.01	0.00	-0.00	0.10	0.32	0.05	0.02	-0.01	0.37
H-L	0.15	0.21	0.24	0.23	0.00	1.56	1.43	1.75	1.63	0.00
China										
H	0.17	0.31	0.31	0.31	0.11	0.44	0.83	0.81	0.82	0.35
L	0.05	-0.01	-0.01	-0.01	0.13	0.16	-0.04	-0.02	-0.04	0.34
H-L	0.12	0.33	0.31	0.33	-0.01	0.58	1.96	1.88	1.95	-0.06
South Korea										
H	0.29	0.52	0.50	0.54	0.03	0.73	1.15	1.24	1.10	0.09
L	-0.13	-0.26	-0.28	-0.26	-0.02	-0.36	-0.73	-0.83	-0.73	-0.04
H-L	0.42	0.78	0.78	0.79	0.05	1.70	2.07	2.76	2.01	0.15

(a) Equal-weighted portfolios

	Mean return					Sharpe ratio				
	Org	PrDf	Return	Sharpe	JT	Org	PrDf	Return	Sharpe	JT
UK										
H	0.10	0.14	0.20	0.15	0.10	0.24	0.80	0.84	1.02	0.47
L	-0.13	-0.15	-0.12	-0.12	-0.08	-0.35	-0.45	-0.39	-0.43	-0.19
H-L	0.23	0.30	0.32	0.27	0.19	0.66	0.91	1.09	1.11	0.43
Japan										
H	0.13	0.11	0.12	0.13	0.05	0.51	0.50	0.49	0.58	0.22
L	0.03	-0.03	-0.03	-0.04	0.03	0.10	-0.13	-0.12	-0.16	0.10
H-L	0.10	0.14	0.15	0.16	0.02	0.62	0.71	0.78	0.85	0.09
China										
H	0.12	0.30	0.28	0.30	0.11	0.31	0.80	0.76	0.80	0.37
L	0.06	-0.01	-0.01	-0.01	0.09	0.20	-0.02	-0.03	-0.03	0.24
H-L	0.05	0.31	0.29	0.31	0.03	0.20	1.31	1.29	1.32	0.10
South Korea										
H	0.18	0.31	0.30	0.32	-0.07	0.44	0.79	0.73	0.82	-0.17
L	-0.14	-0.26	-0.25	-0.22	-0.13	-0.35	-0.73	-0.72	-0.64	-0.27
H-L	0.32	0.57	0.55	0.54	0.06	1.06	2.00	1.88	2.09	0.19

(b) Value-weighted portfolios

Table 7: Factor regression: including size

This table reports the factor regression results of the long (H), short (L), and long-short (H-L) portfolios constructed by DM (Return) including size dummies. The portfolio returns are regressed on the Fama-French three factors (Mkt-Rf, SMB, HML) plus the momentum factor (MOM). For each portfolio, the first row reports the coefficients and the second row reports the Newey-West adjusted t -statistics. The sample period is from 2000 to 2019 except for the UK, for which the sample period ends in 2017 due to data availability.

		α	Mkt-Rf	SMB	HML	MOM	α	Mkt-Rf	SMB	HML	MOM
		Equal-weighted					Value-weighted				
UK	H	0.018	0.064	0.431	-0.170	-0.030	0.014	0.207	0.591	-0.212	-0.042
		3.42	0.48	2.88	-0.99	-0.27	3.00	1.70	4.32	-1.35	-0.41
	L	-0.014	0.108	0.499	-0.203	-0.226	-0.009	0.226	0.647	-0.276	-0.366
		-2.60	0.79	3.23	-1.15	-1.97	-1.44	1.45	3.68	-1.37	-2.80
	H-L	0.030	-0.037	-0.066	0.023	0.194	0.021	-0.012	-0.054	0.053	0.322
		7.35	-0.35	-0.56	0.17	2.24	3.47	-0.07	-0.31	0.27	2.47
Japan	H	0.019	0.081	0.319	-0.190	-0.083	0.008	0.144	0.567	-0.193	-0.136
		4.45	0.87	1.90	-1.26	-1.13	1.65	1.42	3.12	-1.18	-1.71
	L	-0.002	0.125	0.427	-0.118	-0.165	-0.005	0.247	0.326	0.053	-0.076
		-0.40	1.22	2.33	-0.72	-2.06	-1.15	2.36	1.73	0.31	-0.92
	H-L	0.019	-0.039	-0.101	-0.081	0.080	0.012	-0.099	0.248	-0.255	-0.063
		7.35	-0.67	-0.96	-0.86	1.75	3.16	-1.20	1.69	-1.93	-0.98
China	H	0.022	0.085	0.290	-0.050	0.002	0.021	0.057	0.291	-0.052	0.069
		2.87	0.86	1.58	-0.18	0.02	2.81	0.59	1.63	-0.19	0.61
	L	-0.001	0.139	0.168	-0.171	0.078	-0.001	0.164	0.045	-0.275	0.079
		-0.20	1.53	1.00	-0.66	0.74	-0.07	1.88	0.28	-1.10	0.78
	H-L	0.022	-0.053	0.121	0.122	-0.076	0.020	-0.106	0.245	0.223	-0.009
		6.41	-1.21	1.52	0.98	-1.50	4.30	-1.79	2.27	1.32	-0.13
S. Korea	H	0.037	1.434	1.602	-0.657	-0.337	0.020	1.587	1.421	-0.880	-0.236
		7.36	14.97	12.34	-4.59	-1.91	3.68	15.10	9.98	-5.60	-1.22
	L	-0.029	1.341	1.304	-0.579	-0.133	-0.026	1.382	1.067	-0.534	-0.151
		-7.34	17.54	12.59	-5.06	-0.95	-6.28	17.34	9.88	-4.48	-1.03
	H-L	0.063	0.096	0.297	-0.079	-0.206	0.043	0.208	0.353	-0.347	-0.087
		11.71	0.93	2.12	-0.51	-1.08	7.81	1.95	2.45	-2.18	-0.45

to DM, significant improvements are observed across size deciles in most countries. In all countries, the H decile cells reveal darker red colours, which means higher returns, and the L decile cells reveal darker blue colours, which means lower returns. One exception is Japan, where large firms in L tend to have higher returns when size dummies are included. The improvement is especially noticeable in South Korea, where H is dominated by dark red cells and L is dominated by dark blue cells. Consistent with the empirical fact that large firms perform better than small firms in South Korea, we can observe that large firms generally have higher returns than small firms in the same predicted class.

4.6 Performance during the global financial crisis

The momentum strategy is known to crash in the US market when the market recovers from the global financial crisis in 2008. To examine the performance of DM and JT more closely around the financial crisis, we report in Table 8 their performance before (2004-2006), during (2007-2009), and after (2010-2012) the financial crisis. In the table, DM refers to the DM model that includes size dummies and reclassified on Return. As a benchmark, we also include market indexes: FTSE100 for the UK, Nikkei225 for Japan, Shanghai Securities Composite Index for China, and KOSPI for South Korea. Figure 4 presents cumulative returns around the financial crisis.

During the financial crisis, the momentum strategy performs poorly in the UK, Japan, and China. It renders a negative return in these countries and underperforms the market index in the UK and China. The only exception is South Korea, where the momentum strategy performs well with a Sharpe ratio of 1.031. Figure 4 reveals that the momentum strategy behaves rather differently in the four countries during the financial crisis. It performs well in the UK and Japan during the crisis but crashes sharply when the market starts to rebound. In China, it performs poorly throughout the period, whereas in South Korea, it performs steadily without being affected by the financial crisis.

In contrast, the performance of DM during the financial crisis is remarkable. It performs consistently throughout all three periods and performs even better during the crisis in Japan, China, and South Korea. In South Korea, DM achieves a Sharpe ratio of 4.53 during the financial crisis. DM appears to identify losers very well during the crisis and the following recovery period. Figure 4 confirms that DM does not suffer any significant drawdowns during the financial crisis.

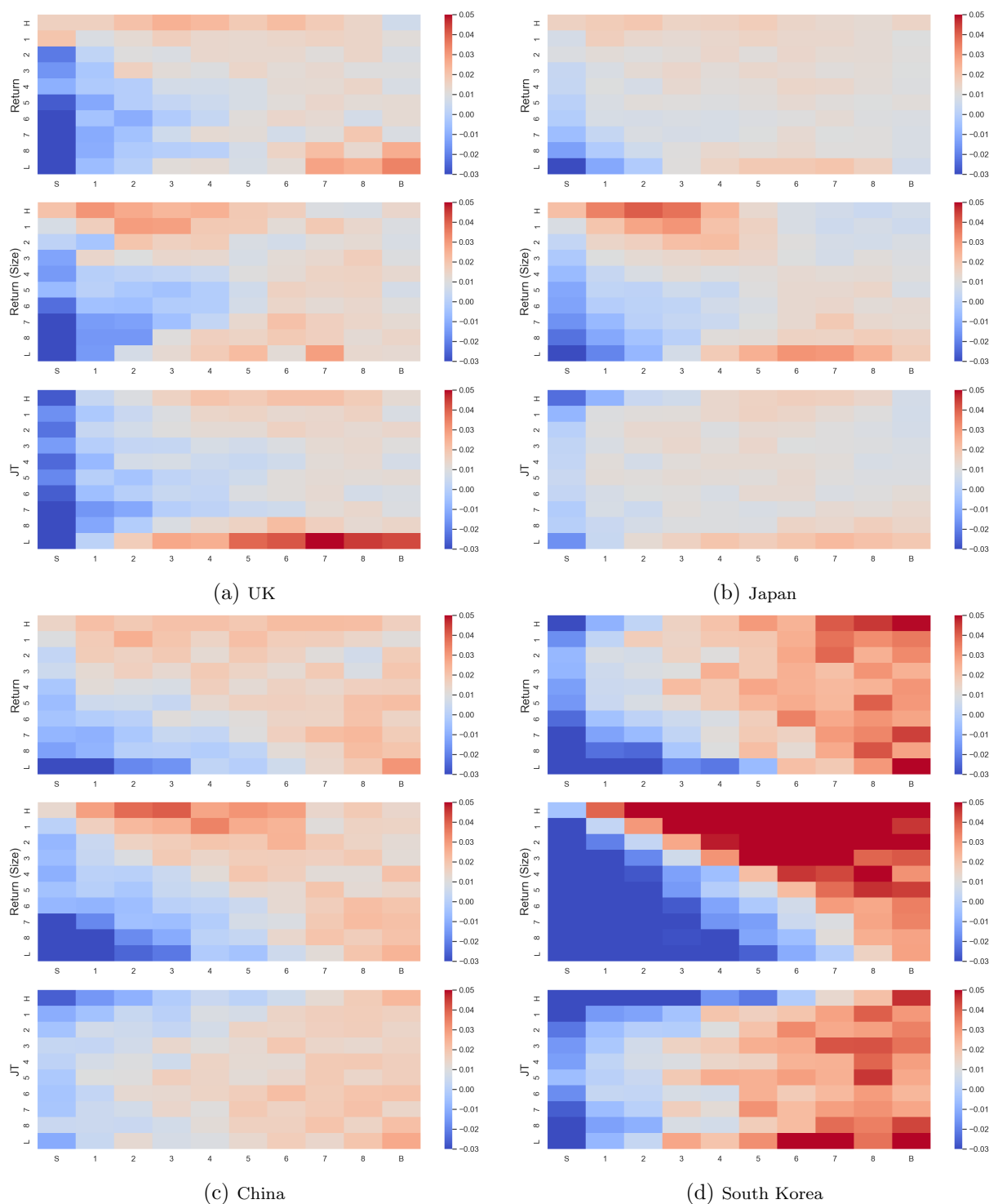


Figure 3: The average return within a predicted class and size decile

This figure presents the average return of the stocks within a predicted class and size decile via heatmaps. The horizontal axis represents the size decile (small to big), the vertical axis represents the predicted class (low to high) by a model, and the colour of a cell represents the average return of the stocks in the cell (red being high and blue being low). 'Return' and 'Return (Size)' on the vertical axis respectively refer to the DM (Return) model with and without size dummies, and 'JT' refers to the traditional momentum strategy.

Han (2022) also finds that DM performs better during the crisis in the US.

Table 8: Performance around the global financial crisis

This table reports the annualized mean returns (Mean), standard deviations (Std), and Sharpe ratios (Sharpe) of the long-short portfolios constructed by DM (Return) including size dummies and the traditional momentum strategy (JT) around the global financial crisis. The benchmark indexes are FTSE100 for the UK, Nikkei225 for Japan, Shanghai Securities Composite Index for China, and KOSPI for South Korea.

	2004.01 - 2006.12			2007.01 - 2009.12			2010.01 - 2012.12		
	Mean	Std	Sharpe	Mean	Std	Sharpe	Mean	Std	Sharpe
UK									
DM	0.400	0.545	2.261	0.381	0.752	1.600	0.405	0.720	1.931
JT	0.328	0.491	1.996	-0.174	1.746	-0.413	0.262	0.865	1.033
Index	0.113	0.254	0.930	-0.030	0.627	-0.347	0.038	0.474	0.244
Japan									
DM	0.253	0.377	2.041	0.349	0.480	2.364	0.255	0.527	1.670
JT	0.221	0.584	1.134	-0.052	1.035	-0.244	0.025	0.429	0.197
Index	0.170	0.491	0.996	-0.131	0.853	-0.616	0.014	0.674	0.067
China									
DM	0.239	0.516	1.456	0.522	0.741	2.290	0.330	0.538	1.942
JT	0.245	1.014	0.761	-0.277	0.879	-1.214	-0.025	0.525	-0.358
Index	0.228	0.922	0.774	0.158	1.447	0.303	-0.102	0.709	-0.638
South Korea									
DM	1.119	0.996	3.757	0.934	0.677	4.531	0.551	0.680	2.651
JT	0.017	0.976	-0.078	0.338	0.986	1.031	0.340	0.988	1.083
Index	0.208	0.631	0.925	0.091	0.946	0.173	0.071	0.573	0.240

5 Conclusion

Han (2022) documents the bimodal distribution of momentum stocks' relative return in the US market, which implies that winners have a high chance of becoming losers in the future and vice versa. He develops a novel machine learning return prediction model, deep momentum, to address bimodality and shows that it significantly outperforms the traditional momentum strategy. We investigate the existence of bimodality and the profitability of deep momentum in a broader region using samples from the UK, Japan, China, and South Korea. Bimodality turns out to be a global phenomenon and momentum exhibits little predictive power in Asian markets. On the contrary, deep momentum effectively addresses bimodality and generates significant profits in these markets. It significantly outperforms the traditional momentum strategy and yields a Sharpe ratio of 1.97, 1.75, 1.88, and 2.76, respectively in the UK, Japan, China, and South Korea. It is remarkable that deep momentum performs consistently well across countries regardless of the performance of

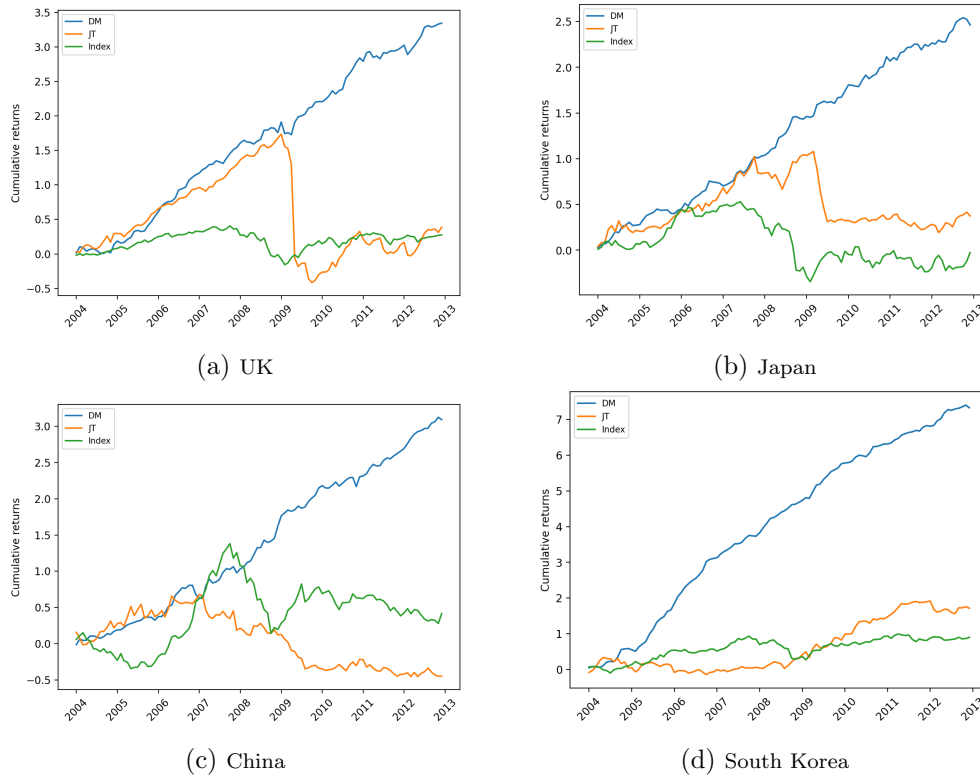


Figure 4: Cumulative Returns around the global financial crisis

This figure presents the cumulative returns (logarithmic scale) of the equal-weighted long-short portfolios constructed by DM (Return) including size dummies (DM) and the traditional momentum strategy (JT), and a market index (Index) around the global financial crisis. The benchmark indexes are FTSE100 for the UK, Nikkei225 for Japan, Shanghai Securities Composite Index for China, and KOSPI for South Korea.

the traditional momentum strategy. The superior performance of deep momentum suggests that even when a single momentum feature cannot predict future returns, several momentum features combined together via a neural network can demonstrate significant predictive power. It will be interesting to expand this research to other countries and asset classes.

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